

4

Linear Methods for Classification

4.1 Introduction

In this chapter we revisit the classification problem and focus on linear methods for classification. Since our predictor $G(x)$ takes values in a discrete set $\mathcal{G}$, we can always divide the input space into a collection of regions labeled according to the classification. We saw in Chapter 2 that the boundaries of these regions can be rough or smooth, depending on the prediction function. For an important class of procedures, these *decision boundaries* are linear; this is what we will mean by linear methods for classification.

There are several different ways in which linear decision boundaries can be found. In Chapter 2 we fit linear regression models to the class indicator variables, and classify to the largest fit. Suppose there are K classes, for convenience labeled $1, 2, \dots, K$, and the fitted linear model for the k th indicator response variable is $\hat{f}_k(x) = \hat{\beta}_{k0} + \hat{\beta}_k^T x$. The decision boundary between class k and ℓ is that set of points for which $\hat{f}_k(x) = \hat{f}_\ell(x)$, that is, the set $\{x : (\hat{\beta}_{k0} - \hat{\beta}_{\ell0}) + (\hat{\beta}_k - \hat{\beta}_\ell)^T x = 0\}$, an affine set or hyperplane¹. Since the same is true for any pair of classes, the input space is divided into regions of constant classification, with piecewise hyperplanar decision boundaries. This regression approach is a member of a class of methods that model *discriminant functions* $\delta_k(x)$ for each class, and then classify x to the class with the largest value for its discriminant function. Methods

¹Strictly speaking, a hyperplane passes through the origin, while an affine set need not. We sometimes ignore the distinction and refer in general to hyperplanes.

that model the posterior probabilities $\Pr(G = k|X = x)$ are also in this class. Clearly, if either the $\delta_k(x)$ or $\Pr(G = k|X = x)$ are linear in x , then the decision boundaries will be linear.

Actually, all we require is that some monotone transformation of δ_k or $\Pr(G = k|X = x)$ be linear for the decision boundaries to be linear. For example, if there are two classes, a popular model for the posterior probabilities is

$$\begin{aligned}\Pr(G = 1|X = x) &= \frac{\exp(\beta_0 + \beta^T x)}{1 + \exp(\beta_0 + \beta^T x)}, \\ \Pr(G = 2|X = x) &= \frac{1}{1 + \exp(\beta_0 + \beta^T x)}.\end{aligned}\tag{4.1}$$

Here the monotone transformation is the *logit* transformation: $\log[p/(1-p)]$, and in fact we see that

$$\log \frac{\Pr(G = 1|X = x)}{\Pr(G = 2|X = x)} = \beta_0 + \beta^T x.\tag{4.2}$$

The decision boundary is the set of points for which the *log-odds* are zero, and this is a hyperplane defined by $\{x|\beta_0 + \beta^T x = 0\}$. We discuss two very popular but different methods that result in linear log-odds or logits: linear discriminant analysis and linear logistic regression. Although they differ in their derivation, the essential difference between them is in the way the linear function is fit to the training data.

A more direct approach is to explicitly model the boundaries between the classes as linear. For a two-class problem in a p -dimensional input space, this amounts to modeling the decision boundary as a hyperplane—in other words, a normal vector and a cut-point. We will look at two methods that explicitly look for “separating hyperplanes.” The first is the well-known *perceptron* model of Rosenblatt (1958), with an algorithm that finds a separating hyperplane in the training data, if one exists. The second method, due to Vapnik (1996), finds an *optimally separating hyperplane* if one exists, else finds a hyperplane that minimizes some measure of overlap in the training data. We treat the separable case here, and defer treatment of the nonseparable case to Chapter 12.

While this entire chapter is devoted to linear decision boundaries, there is considerable scope for generalization. For example, we can expand our variable set $X_1, \dots, X_p$ by including their squares and cross-products $X_1^2, X_2^2, \dots, X_1X_2, \dots$, thereby adding $p(p+1)/2$ additional variables. Linear functions in the augmented space map down to quadratic functions in the original space—hence linear decision boundaries to quadratic decision boundaries. Figure 4.1 illustrates the idea. The data are the same: the left plot uses linear decision boundaries in the two-dimensional space shown, while the right plot uses linear decision boundaries in the augmented five-dimensional space described above. This approach can be used with any basis transfor-

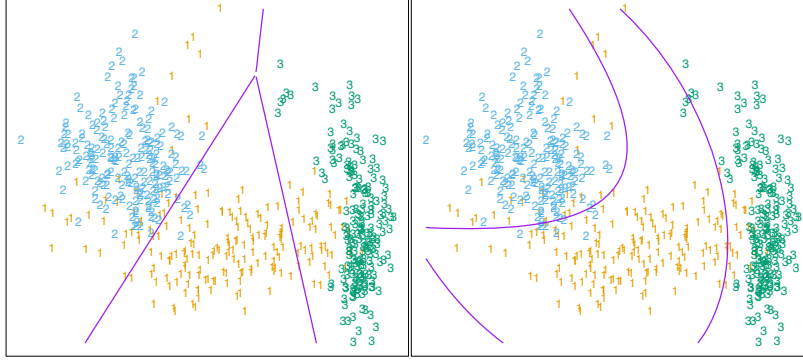


FIGURE 4.1. The left plot shows some data from three classes, with linear decision boundaries found by linear discriminant analysis. The right plot shows quadratic decision boundaries. These were obtained by finding linear boundaries in the five-dimensional space $X_1, X_2, X_1X_2, X_1^2, X_2^2$. Linear inequalities in this space are quadratic inequalities in the original space.

mation $h(X)$ where $h : \mathbb{R}^p \mapsto \mathbb{R}^q$ with $q > p$, and will be explored in later chapters.

4.2 Linear Regression of an Indicator Matrix

Here each of the response categories are coded via an indicator variable. Thus if $\mathcal{G}$ has K classes, there will be K such indicators Y_k , $k = 1, \dots, K$, with $Y_k = 1$ if $G = k$ else 0. These are collected together in a vector $Y = (Y_1, \dots, Y_K)$, and the N training instances of these form an $N \times K$ indicator response matrix $\mathbf{Y}$. $\mathbf{Y}$ is a matrix of 0's and 1's, with each row having a single 1. We fit a linear regression model to each of the columns of $\mathbf{Y}$ simultaneously, and the fit is given by

$$\hat{\mathbf{Y}} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y}. \quad (4.3)$$

Chapter 3 has more details on linear regression. Note that we have a coefficient vector for each response column $\mathbf{y}_k$, and hence a $(p+1) \times K$ coefficient matrix $\hat{\mathbf{B}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y}$. Here $\mathbf{X}$ is the model matrix with $p+1$ columns corresponding to the p inputs, and a leading column of 1's for the intercept.

A new observation with input x is classified as follows:

- compute the fitted output $\hat{f}(x) = [(1, x)\hat{\mathbf{B}}]^T$, a K vector;
- identify the largest component and classify accordingly:

$$\hat{G}(x) = \operatorname{argmax}_{k \in \mathcal{G}} \hat{f}_k(x). \quad (4.4)$$

What is the rationale for this approach? One rather formal justification is to view the regression as an estimate of conditional expectation. For the random variable Y_k , $E(Y_k|X = x) = \Pr(G = k|X = x)$, so conditional expectation of each of the Y_k seems a sensible goal. The real issue is: how good an approximation to conditional expectation is the rather rigid linear regression model? Alternatively, are the $\hat{f}_k(x)$ reasonable estimates of the posterior probabilities $\Pr(G = k|X = x)$, and more importantly, does this matter?

It is quite straightforward to verify that $\sum_{k \in \mathcal{G}} \hat{f}_k(x) = 1$ for any x , as long as there is an intercept in the model (column of 1's in $\mathbf{X}$). However, the $\hat{f}_k(x)$ can be negative or greater than 1, and typically some are. This is a consequence of the rigid nature of linear regression, especially if we make predictions outside the hull of the training data. These violations in themselves do not guarantee that this approach will not work, and in fact on many problems it gives similar results to more standard linear methods for classification. If we allow linear regression onto basis expansions $h(X)$ of the inputs, this approach can lead to consistent estimates of the probabilities. As the size of the training set N grows bigger, we adaptively include more basis elements so that linear regression onto these basis functions approaches conditional expectation. We discuss such approaches in Chapter 5.

A more simplistic viewpoint is to construct *targets* t_k for each class, where t_k is the k th column of the $K \times K$ identity matrix. Our prediction problem is to try and reproduce the appropriate target for an observation. With the same coding as before, the response vector y_i (i th row of $\mathbf{Y}$) for observation i has the value $y_i = t_k$ if $g_i = k$. We might then fit the linear model by least squares:

$$\min_{\mathbf{B}} \sum_{i=1}^N \|y_i - [(1, x_i)\mathbf{B}]^T\|^2. \quad (4.5)$$

The criterion is a sum-of-squared Euclidean distances of the fitted vectors from their targets. A new observation is classified by computing its fitted vector $\hat{f}(x)$ and classifying to the closest target:

$$\hat{G}(x) = \operatorname{argmin}_k \|\hat{f}(x) - t_k\|^2. \quad (4.6)$$

This is exactly the same as the previous approach:

- The sum-of-squared-norm criterion is exactly the criterion for multiple response linear regression, just viewed slightly differently. Since a squared norm is itself a sum of squares, the components decouple and can be rearranged as a separate linear model for each element. Note that this is only possible because there is nothing in the model that binds the different responses together.

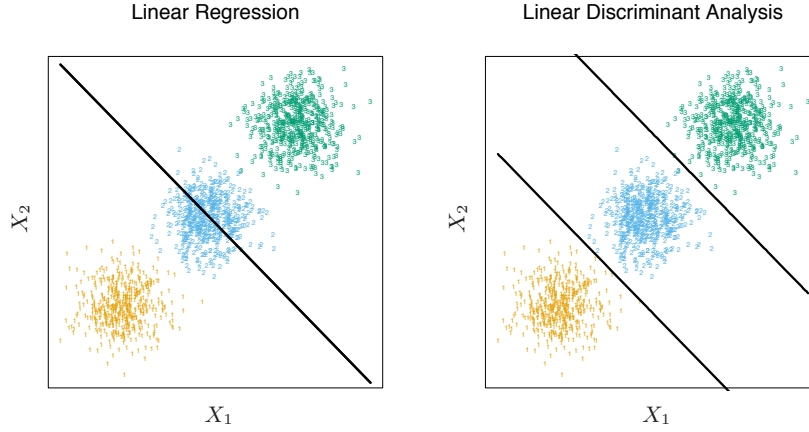


FIGURE 4.2. The data come from three classes in $\mathbb{R}^2$ and are easily separated by linear decision boundaries. The right plot shows the boundaries found by linear discriminant analysis. The left plot shows the boundaries found by linear regression of the indicator response variables. The middle class is completely masked (never dominates).

- The closest target classification rule (4.6) is easily seen to be exactly the same as the maximum fitted component criterion (4.4), but does require that the fitted values sum to 1.

There is a serious problem with the regression approach when the number of classes $K \geq 3$, especially prevalent when K is large. Because of the rigid nature of the regression model, classes can be *masked* by others. Figure 4.2 illustrates an extreme situation when $K = 3$. The three classes are perfectly separated by linear decision boundaries, yet linear regression misses the middle class completely.

In Figure 4.3 we have projected the data onto the line joining the three centroids (there is no information in the orthogonal direction in this case), and we have included and coded the three response variables Y_1 , Y_2 and Y_3 . The three regression lines (left panel) are included, and we see that the line corresponding to the middle class is horizontal and its fitted values are never dominant! Thus, observations from class 2 are classified either as class 1 or class 3. The right panel uses quadratic regression rather than linear regression. For this simple example a quadratic rather than linear fit (for the middle class at least) would solve the problem. However, it can be seen that if there were four rather than three classes lined up like this, a quadratic would not come down fast enough, and a cubic would be needed as well. A loose but general rule is that if $K \geq 3$ classes are lined up, polynomial terms up to degree $K - 1$ might be needed to resolve them. Note also that these are polynomials along the derived direction

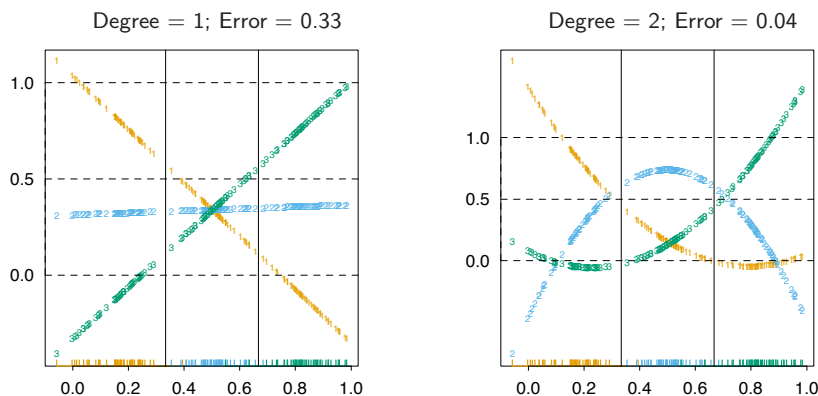


FIGURE 4.3. The effects of masking on linear regression in $\mathbb{R}^2$ for a three-class problem. The rug plot at the base indicates the positions and class membership of each observation. The three curves in each panel are the fitted regressions to the three-class indicator variables; for example, for the blue class, y_{blue} is 1 for the blue observations, and 0 for the green and orange. The fits are linear and quadratic polynomials. Above each plot is the training error rate. The Bayes error rate is 0.025 for this problem, as is the LDA error rate.

passing through the centroids, which can have arbitrary orientation. So in p -dimensional input space, one would need general polynomial terms and cross-products of total degree $K - 1$, $O(p^{K-1})$ terms in all, to resolve such worst-case scenarios.

The example is extreme, but for large K and small p such maskings naturally occur. As a more realistic illustration, Figure 4.4 is a projection of the training data for a vowel recognition problem onto an informative two-dimensional subspace. There are $K = 11$ classes in $p = 10$ dimensions. This is a difficult classification problem, and the best methods achieve around 40% errors on the test data. The main point here is summarized in Table 4.1; linear regression has an error rate of 67%, while a close relative, linear discriminant analysis, has an error rate of 56%. It seems that masking has hurt in this case. While all the other methods in this chapter are based on linear functions of x as well, they use them in such a way that avoids this masking problem.

4.3 Linear Discriminant Analysis

Decision theory for classification (Section 2.4) tells us that we need to know the class posteriors $\Pr(G|X)$ for optimal classification. Suppose $f_k(x)$ is the class-conditional density of X in class $G = k$, and let π_k be the prior probability of class k , with $\sum_{k=1}^K \pi_k = 1$. A simple application of Bayes

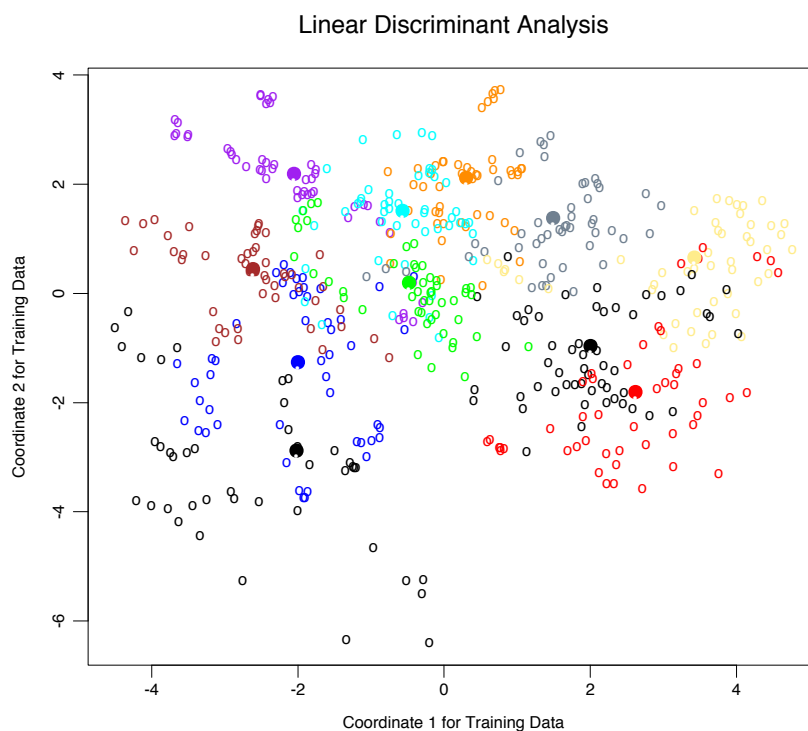


FIGURE 4.4. A two-dimensional plot of the vowel training data. There are eleven classes with $X \in \mathbb{R}^{10}$, and this is the best view in terms of a LDA model (Section 4.3.3). The heavy circles are the projected mean vectors for each class. The class overlap is considerable.

TABLE 4.1. Training and test error rates using a variety of linear techniques on the vowel data. There are eleven classes in ten dimensions, of which three account for 90% of the variance (via a principal components analysis). We see that linear regression is hurt by masking, increasing the test and training error by over 10%.

Technique	Error Rates	
	Training	Test
Linear regression	0.48	0.67
Linear discriminant analysis	0.32	0.56
Quadratic discriminant analysis	0.01	0.53
Logistic regression	0.22	0.51

theorem gives us

$$\Pr(G = k|X = x) = \frac{f_k(x)\pi_k}{\sum_{\ell=1}^K f_\ell(x)\pi_\ell}. \quad (4.7)$$

We see that in terms of ability to classify, having the $f_k(x)$ is almost equivalent to having the quantity $\Pr(G = k|X = x)$.

Many techniques are based on models for the class densities:

- linear and quadratic discriminant analysis use Gaussian densities;
- more flexible mixtures of Gaussians allow for nonlinear decision boundaries (Section 6.8);
- general nonparametric density estimates for each class density allow the most flexibility (Section 6.6.2);
- *Naive Bayes* models are a variant of the previous case, and assume that each of the class densities are products of marginal densities; that is, they assume that the inputs are conditionally independent in each class (Section 6.6.3).

Suppose that we model each class density as multivariate Gaussian

$$f_k(x) = \frac{1}{(2\pi)^{p/2}|\Sigma_k|^{1/2}} e^{-\frac{1}{2}(x-\mu_k)^T \Sigma_k^{-1}(x-\mu_k)}. \quad (4.8)$$

Linear discriminant analysis (LDA) arises in the special case when we assume that the classes have a common covariance matrix $\Sigma_k = \Sigma \forall k$. In comparing two classes k and ℓ , it is sufficient to look at the log-ratio, and we see that

$$\begin{aligned} \log \frac{\Pr(G = k|X = x)}{\Pr(G = \ell|X = x)} &= \log \frac{f_k(x)}{f_\ell(x)} + \log \frac{\pi_k}{\pi_\ell} \\ &= \log \frac{\pi_k}{\pi_\ell} - \frac{1}{2}(\mu_k + \mu_\ell)^T \Sigma^{-1}(\mu_k - \mu_\ell) \\ &\quad + x^T \Sigma^{-1}(\mu_k - \mu_\ell), \end{aligned} \quad (4.9)$$

an equation linear in x . The equal covariance matrices cause the normalization factors to cancel, as well as the quadratic part in the exponents. This linear log-odds function implies that the decision boundary between classes k and ℓ —the set where $\Pr(G = k|X = x) = \Pr(G = \ell|X = x)$ —is linear in x ; in p dimensions a hyperplane. This is of course true for any pair of classes, so all the decision boundaries are linear. If we divide $\mathbb{R}^p$ into regions that are classified as class 1, class 2, etc., these regions will be separated by hyperplanes. Figure 4.5 (left panel) shows an idealized example with three classes and $p = 2$. Here the data do arise from three Gaussian distributions with a common covariance matrix. We have included in

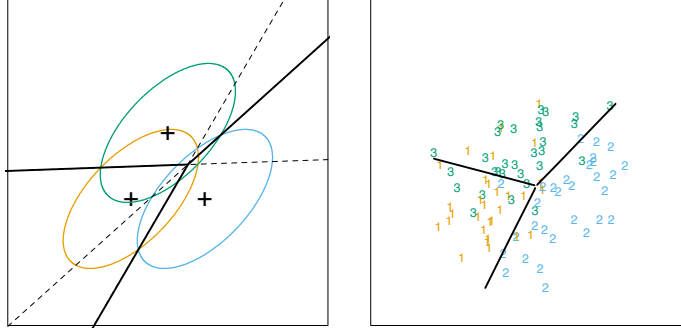


FIGURE 4.5. The left panel shows three Gaussian distributions, with the same covariance and different means. Included are the contours of constant density enclosing 95% of the probability in each case. The Bayes decision boundaries between each pair of classes are shown (broken straight lines), and the Bayes decision boundaries separating all three classes are the thicker solid lines (a subset of the former). On the right we see a sample of 30 drawn from each Gaussian distribution, and the fitted LDA decision boundaries.

the figure the contours corresponding to 95% highest probability density, as well as the class centroids. Notice that the decision boundaries are not the perpendicular bisectors of the line segments joining the centroids. This would be the case if the covariance Σ were spherical $\sigma^2 \mathbf{I}$, and the class priors were equal. From (4.9) we see that the *linear discriminant functions*

$$\delta_k(x) = x^T \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + \log \pi_k \quad (4.10)$$

are an equivalent description of the decision rule, with $G(x) = \operatorname{argmax}_k \delta_k(x)$.

In practice we do not know the parameters of the Gaussian distributions, and will need to estimate them using our training data:

- $\hat{\pi}_k = N_k/N$, where N_k is the number of class- k observations;
- $\hat{\mu}_k = \sum_{g_i=k} x_i / N_k$;
- $\hat{\Sigma} = \sum_{k=1}^K \sum_{g_i=k} (x_i - \hat{\mu}_k)(x_i - \hat{\mu}_k)^T / (N - K)$.

Figure 4.5 (right panel) shows the estimated decision boundaries based on a sample of size 30 each from three Gaussian distributions. Figure 4.1 on page 103 is another example, but here the classes are not Gaussian.

With two classes there is a simple correspondence between linear discriminant analysis and classification by linear least squares, as in (4.5). The LDA rule classifies to class 2 if

$$x^T \hat{\Sigma}^{-1} (\hat{\mu}_2 - \hat{\mu}_1) > \frac{1}{2} \hat{\mu}_2^T \hat{\Sigma}^{-1} \hat{\mu}_2 - \frac{1}{2} \hat{\mu}_1^T \hat{\Sigma}^{-1} \hat{\mu}_1 + \log(N_1/N) - \log(N_2/N) \quad (4.11)$$

and class 1 otherwise. Suppose we code the targets in the two classes as $+1$ and -1 , respectively. It is easy to show that the coefficient vector from least squares is proportional to the LDA direction given in (4.11) (Exercise 4.2). [In fact, this correspondence occurs for any (distinct) coding of the targets; see Exercise 4.2]. However unless $N_1 = N_2$ the intercepts are different and hence the resulting decision rules are different.

Since this derivation of the LDA direction via least squares does not use a Gaussian assumption for the features, its applicability extends beyond the realm of Gaussian data. However the derivation of the particular intercept or cut-point given in (4.11) *does* require Gaussian data. Thus it makes sense to instead choose the cut-point that empirically minimizes training error for a given dataset. This is something we have found to work well in practice, but have not seen it mentioned in the literature.

With more than two classes, LDA is not the same as linear regression of the class indicator matrix, and it avoids the masking problems associated with that approach (Hastie et al., 1994). A correspondence between regression and LDA can be established through the notion of *optimal scoring*, discussed in Section 12.5.

Getting back to the general discriminant problem (4.8), if the Σ_k are not assumed to be equal, then the convenient cancellations in (4.9) do not occur; in particular the pieces quadratic in x remain. We then get *quadratic discriminant functions* (QDA),

$$\delta_k(x) = -\frac{1}{2} \log |\Sigma_k| - \frac{1}{2} (x - \mu_k)^T \Sigma_k^{-1} (x - \mu_k) + \log \pi_k. \quad (4.12)$$

The decision boundary between each pair of classes k and ℓ is described by a quadratic equation $\{x : \delta_k(x) = \delta_\ell(x)\}$.

Figure 4.6 shows an example (from Figure 4.1 on page 103) where the three classes are Gaussian mixtures (Section 6.8) and the decision boundaries are approximated by quadratic equations in x . Here we illustrate two popular ways of fitting these quadratic boundaries. The right plot uses QDA as described here, while the left plot uses LDA in the enlarged five-dimensional quadratic polynomial space. The differences are generally small; QDA is the preferred approach, with the LDA method a convenient substitute ².

The estimates for QDA are similar to those for LDA, except that separate covariance matrices must be estimated for each class. When p is large this can mean a dramatic increase in parameters. Since the decision boundaries are functions of the parameters of the densities, counting the number of parameters must be done with care. For LDA, it seems there are $(K - 1) \times (p + 1)$ parameters, since we only need the differences $\delta_k(x) - \delta_K(x)$

²For this figure and many similar figures in the book we compute the decision boundaries by an exhaustive contouring method. We compute the decision rule on a fine lattice of points, and then use contouring algorithms to compute the boundaries.

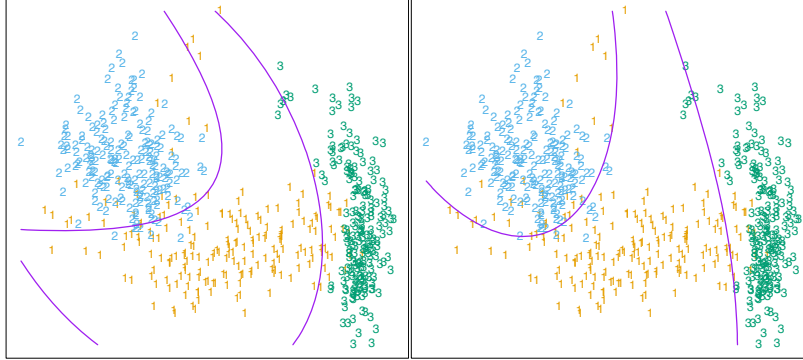


FIGURE 4.6. Two methods for fitting quadratic boundaries. The left plot shows the quadratic decision boundaries for the data in Figure 4.1 (obtained using LDA in the five-dimensional space $X_1, X_2, X_1X_2, X_1^2, X_2^2$). The right plot shows the quadratic decision boundaries found by QDA. The differences are small, as is usually the case.

between the discriminant functions where K is some pre-chosen class (here we have chosen the last), and each difference requires $p + 1$ parameters³. Likewise for QDA there will be $(K - 1) \times \{p(p + 3)/2 + 1\}$ parameters. Both LDA and QDA perform well on an amazingly large and diverse set of classification tasks. For example, in the STATLOG project (Michie et al., 1994) LDA was among the top three classifiers for 7 of the 22 datasets, QDA among the top three for four datasets, and one of the pair were in the top three for 10 datasets. Both techniques are widely used, and entire books are devoted to LDA. It seems that whatever exotic tools are the rage of the day, we should always have available these two simple tools. The question arises why LDA and QDA have such a good track record. The reason is not likely to be that the data are approximately Gaussian, and in addition for LDA that the covariances are approximately equal. More likely a reason is that the data can only support simple decision boundaries such as linear or quadratic, and the estimates provided via the Gaussian models are stable. This is a bias variance tradeoff—we can put up with the bias of a linear decision boundary because it can be estimated with much lower variance than more exotic alternatives. This argument is less believable for QDA, since it can have many parameters itself, although perhaps fewer than the non-parametric alternatives.

³Although we fit the covariance matrix $\hat{\Sigma}$ to compute the LDA discriminant functions, a much reduced function of it is all that is required to estimate the $O(p)$ parameters needed to compute the decision boundaries.

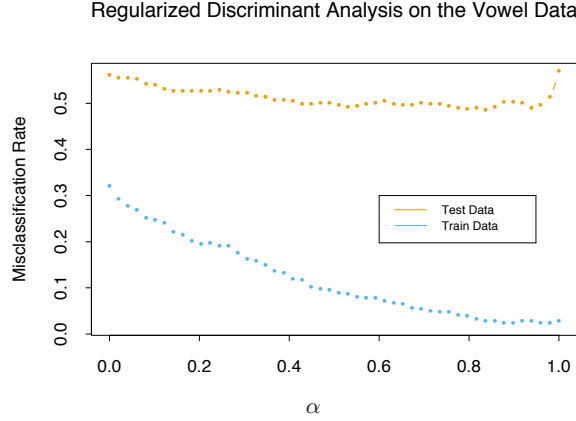


FIGURE 4.7. Test and training errors for the vowel data, using regularized discriminant analysis with a series of values of $\alpha \in [0, 1]$. The optimum for the test data occurs around $\alpha = 0.9$, close to quadratic discriminant analysis.

4.3.1 Regularized Discriminant Analysis

Friedman (1989) proposed a compromise between LDA and QDA, which allows one to shrink the separate covariances of QDA toward a common covariance as in LDA. These methods are very similar in flavor to ridge regression. The regularized covariance matrices have the form

$$\hat{\Sigma}_k(\alpha) = \alpha \hat{\Sigma}_k + (1 - \alpha) \hat{\Sigma}, \quad (4.13)$$

where $\hat{\Sigma}$ is the pooled covariance matrix as used in LDA. Here $\alpha \in [0, 1]$ allows a continuum of models between LDA and QDA, and needs to be specified. In practice α can be chosen based on the performance of the model on validation data, or by cross-validation.

Figure 4.7 shows the results of RDA applied to the vowel data. Both the training and test error improve with increasing α , although the test error increases sharply after $\alpha = 0.9$. The large discrepancy between the training and test error is partly due to the fact that there are many repeat measurements on a small number of individuals, different in the training and test set.

Similar modifications allow $\hat{\Sigma}$ itself to be shrunk toward the scalar covariance,

$$\hat{\Sigma}(\gamma) = \gamma \hat{\Sigma} + (1 - \gamma) \hat{\sigma}^2 \mathbf{I} \quad (4.14)$$

for $\gamma \in [0, 1]$. Replacing $\hat{\Sigma}$ in (4.13) by $\hat{\Sigma}(\gamma)$ leads to a more general family of covariances $\hat{\Sigma}(\alpha, \gamma)$ indexed by a pair of parameters.

In Chapter 12, we discuss other regularized versions of LDA, which are more suitable when the data arise from digitized analog signals and images.

In these situations the features are high-dimensional and correlated, and the LDA coefficients can be regularized to be smooth or sparse in the original domain of the signal. This leads to better generalization and allows for easier interpretation of the coefficients. In Chapter 18 we also deal with very high-dimensional problems, where for example the features are gene-expression measurements in microarray studies. There the methods focus on the case $\gamma = 0$ in (4.14), and other severely regularized versions of LDA.

4.3.2 Computations for LDA

As a lead-in to the next topic, we briefly digress on the computations required for LDA and especially QDA. Their computations are simplified by diagonalizing $\hat{\Sigma}$ or $\hat{\Sigma}_k$. For the latter, suppose we compute the eigen-decomposition for each $\hat{\Sigma}_k = \mathbf{U}_k \mathbf{D}_k \mathbf{U}_k^T$, where $\mathbf{U}_k$ is $p \times p$ orthonormal, and $\mathbf{D}_k$ a diagonal matrix of positive eigenvalues $d_{k\ell}$. Then the ingredients for $\delta_k(x)$ (4.12) are

- $(x - \hat{\mu}_k)^T \hat{\Sigma}_k^{-1} (x - \hat{\mu}_k) = [\mathbf{U}_k^T (x - \hat{\mu}_k)]^T \mathbf{D}_k^{-1} [\mathbf{U}_k^T (x - \hat{\mu}_k)];$
- $\log |\hat{\Sigma}_k| = \sum_{\ell} \log d_{k\ell}.$

In light of the computational steps outlined above, the LDA classifier can be implemented by the following pair of steps:

- *Sphere* the data with respect to the common covariance estimate $\hat{\Sigma}$: $X^* \leftarrow \mathbf{D}^{-\frac{1}{2}} \mathbf{U}^T X$, where $\hat{\Sigma} = \mathbf{U} \mathbf{D} \mathbf{U}^T$. The common covariance estimate of X^* will now be the identity.
- Classify to the closest class centroid in the transformed space, modulo the effect of the class prior probabilities π_k .

4.3.3 Reduced-Rank Linear Discriminant Analysis

So far we have discussed LDA as a restricted Gaussian classifier. Part of its popularity is due to an additional restriction that allows us to view informative low-dimensional projections of the data.

The K centroids in p -dimensional input space lie in an affine subspace of dimension $\leq K - 1$, and if p is much larger than K , this will be a considerable drop in dimension. Moreover, in locating the closest centroid, we can ignore distances orthogonal to this subspace, since they will contribute equally to each class. Thus we might just as well project the X^* onto this centroid-spanning subspace H_{K-1} , and make distance comparisons there. Thus there is a fundamental dimension reduction in LDA, namely, that we need only consider the data in a subspace of dimension at most $K - 1$.